

Improving Network Security with Machine Learning: Anomaly Detection using Random Forest, XGBoost, LightGBM, and Extra Trees

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Abstract—Anomaly Detection Systems (ADS) play a crucial role in safeguarding modern network environments against cyberattacks. Traditional rule-based ADS struggle to effectively address evolving and zero-day attacks. Machine Learning (ML) offers a promising alternative by learning from network traffic patterns and classifying them as normal or anomalous. This paper presents a comprehensive study on developing an ML-based ADS using the ADS2018 dataset. We emphasize a rigorous approach to data preprocessing, feature selection, and model optimization. Several prominent classifiers, including Random Forest, XGBoost, LightGBM, and Extra Trees, are evaluated to identify the most effective model. Our findings demonstrate that LightGBM achieves superior performance, attaining a test accuracy of 91.04%, highlighting its potential for robust network anomaly detection. The paper provides detailed insights into the data preprocessing steps, feature selection process, model training methodology, evaluation metrics employed, and a comparative analysis of the classifiers, supplemented with visualizations to enhance understanding of model behavior.

Keywords—Anomaly Detection, Machine Learning, Network Security, ADS2018 Dataset, Random Forest, XGBoost, LightGBM, Extra Trees, Classification, Cybersecurity

I. INTRODUCTION

The rapid growth of internet usage has led to a corresponding surge in cyber threats, making network security a paramount concern for both individuals and organizations. Cyberattacks are becoming increasingly sophisticated, with attackers constantly developing novel methods to exploit vulnerabilities. Traditional Anomaly Detection Systems (ADS), often signature-based, rely on matching predefined patterns to identify attacks. While effective against known threats, these systems struggle with new and emerging attacks, as they require frequent updates to their attack signature database. Anomaly-based ADS offer an alternative by attempting to identify deviations from normal network behavior. However, traditional anomaly-based approaches can suffer from high false positive rates, limiting their practical applicability. Ma-

chine learning (ML) has emerged as a powerful technique to enhance anomaly detection capabilities. Unlike rule-based ADS, ML models can learn complex patterns from historical network traffic and dynamically identify threats without relying on pre-defined signatures. By leveraging large datasets, such as the ADS2018 dataset, ML-based ADS can effectively classify network traffic into normal and malicious categories. Crucially, these models can adapt to new attack patterns, offering improved resilience against unknown threats. However, implementing ML for ADS involves addressing challenges such as data preprocessing, feature selection, class imbalance, and computational complexity. This paper investigates the application of ML algorithms for anomaly detection using the ADS2018 dataset. Our methodology encompasses a comprehensive data preprocessing stage, including handling missing values, encoding categorical features, and normalizing numerical attributes. Feature selection techniques are employed to reduce data dimensionality and optimize model efficiency. We train and evaluate several classification algorithms, specifically Random Forest, XGBoost, LightGBM, and Extra Trees, using key performance metrics such as accuracy, precision, recall, and F1-score. The results presented in this paper represent ongoing work toward the development of efficient and scalable ML-based anomaly detection systems.

II. RELATED WORK

Anomaly detection has been a subject of extensive research, with various approaches proposed over the years. Traditional signature-based ADS, as discussed earlier, suffer from limitations in detecting novel attacks [1]. Anomaly-based detection, while aiming to identify deviations from normal behavior, often struggles with high false positive rates [2]. These shortcomings have motivated the exploration of machine learning techniques for enhanced anomaly detection.

Several studies have demonstrated the effectiveness of machine learning in improving ADS performance. Supervised learning algorithms, in particular, have shown promise in classifying network traffic into normal and attack categories. Random Forest, known for its robustness and ability to handle high-dimensional data, has been successfully applied in anomaly detection [3]–[5]. Gradient boosting algorithms, such as XGBoost and LightGBM, have also gained popularity due to their high accuracy and efficiency [6]–[8]. These algorithms have been shown to outperform traditional machine learning methods in various network security applications. Extra Trees, another ensemble method, offers similar advantages to Random Forest with potentially improved performance in certain scenarios [9], [10]. Deep learning techniques have also been explored for anomaly detection, offering the potential to learn complex patterns from large datasets [11], [12].

Feature selection plays a crucial role in enhancing the performance of ML-based ADS. Techniques like Recursive Feature Elimination (RFE) have been employed to identify the most relevant features and reduce data dimensionality [13], [14]. This not only improves model efficiency but can also contribute to better generalization performance. Information gain and chi-squared tests are also commonly used for feature selection [15].

Data preprocessing is another critical aspect of building effective ML-based ADS. Handling missing values, encoding categorical features, and normalizing numerical attributes are essential steps to ensure data quality and improve the performance of learning algorithms [16], [17]. Addressing class imbalance, a common issue in anomaly detection datasets, is also crucial to prevent biased models [18], [19]. Techniques such as oversampling the minority class (SMOTE) or under-sampling the majority class can be employed to mitigate this problem.

Evaluation metrics are essential for assessing the performance of ADS models. Accuracy, precision, recall, and F1-score are commonly used metrics [20]. ROC curves and AUC are also valuable tools for evaluating the trade-off between true positive rate and false positive rate.

This paper builds upon the existing literature by conducting a comprehensive evaluation of several state-of-the-art machine learning algorithms, including Random Forest, XGBoost, LightGBM, and Extra Trees, for anomaly detection using the ADS2018 dataset. We emphasize a rigorous approach to data preprocessing, feature selection, and model optimization to develop an effective and scalable ML-based ADS.

III. PROBLEM IDENTIFICATION & OBJECTIVE

This section outlines the key challenges addressed in developing an effective machine learning-based anomaly detection system and articulates the specific objectives of this research.

A. Challenges in Anomaly Detection

- **High-Dimensional Data:** Network traffic data is characterized by a high number of features, which can lead to increased computational complexity and potentially

degrade model performance. Effective feature selection is crucial to identify the most relevant features and improve model efficiency.

- **Class Imbalance:** Real-world network traffic datasets often exhibit significant class imbalance, where some attack categories have far fewer instances than normal traffic or other attack types. This imbalance can bias learning algorithms, leading to poor performance on minority classes. Resampling or other techniques are necessary to address this issue.
- **Real-Time Detection:** For practical deployment, an ADS must be capable of classifying threats with minimal latency. This necessitates the development of models that are not only accurate but also computationally efficient.
- **Optimizing Model Selection:** Choosing the optimal model from a range of candidate classifiers is a critical task. A systematic evaluation and comparison of different models are required to identify the best performer for the specific application.

B. Objectives

- **Implement comprehensive preprocessing steps:** This includes feature selection to reduce data dimensionality and improve model performance, as well as class balancing techniques to address the inherent imbalance in the dataset.
- **Train and compare multiple machine learning models:** We will train and evaluate a variety of ML algorithms, including Random Forest, XGBoost, LightGBM, and Extra Trees, to assess their suitability for anomaly detection.
- **Assess performance using relevant metrics:** The performance of the trained models will be evaluated using a comprehensive set of metrics, including accuracy, precision, recall, and F1-score, to provide a holistic view of their effectiveness.
- **Select the best-performing model for potential real-time ADS implementation:** Based on the evaluation results, we will identify the model that offers the best balance of performance and computational efficiency for potential deployment in a real-time anomaly detection system.

IV. EXISTING SYSTEM

Traditional Anomaly Detection Systems (ADS) are broadly categorized into:

A. Signature-Based Detection

This approach relies on matching network traffic against a database of pre-defined attack signatures. While highly effective at identifying known attacks, it is inherently limited in its ability to detect novel or evolving attack patterns (zero-day attacks). Maintaining the effectiveness of signature-based ADS requires frequent updates to the signature database, making it a knowledge-intensive and resource-intensive endeavor. Examples of signature-based ADS include Snort and Suricata, which employ rule sets for anomaly detection.

1) Limitations of Signature-Based ADS:

- **Ineffective against new or polymorphic attacks:** Signature-based systems struggle to recognize variations of known attacks or completely new attack vectors.
- **Maintenance-intensive:** The need for regular updates to the attack signature database creates a significant maintenance burden.
- **Difficulty adapting to changing network environments:** As network environments and attack patterns evolve, signature-based ADS face challenges in remaining effective.

These limitations highlight the shortcomings of traditional ADS solutions in providing comprehensive security against the dynamic landscape of cyber threats. This necessitates the development of advanced ADS models that leverage machine learning techniques to achieve improved adaptability, robustness, and real-time threat detection capabilities.

V. PROPOSED SYSTEM

To overcome the limitations of traditional anomaly detection systems, we propose a machine learning-driven ADS. Our proposed system incorporates the following key components:

A. Data Preprocessing

A crucial initial step involves preprocessing the raw network traffic data. This includes handling missing values, encoding categorical features into numerical representations suitable for ML algorithms, and normalizing numerical features to a standard scale. These preprocessing steps are essential for improving the quality of the data and enhancing the performance of the subsequent learning models.

B. Feature Selection

Given the high dimensionality of network traffic data, feature selection plays a vital role. We employ feature selection techniques to identify and remove redundant or irrelevant attributes, thereby reducing data dimensionality, optimizing model efficiency, and potentially improving model generalization.

C. Training Multiple Classifiers

We train and evaluate several prominent machine learning classifiers to identify the most effective model for anomaly detection. The classifiers considered in this study include Random Forest, XGBoost, LightGBM, and Extra Trees. These algorithms were chosen for their demonstrated effectiveness in various classification tasks and their suitability for handling complex datasets.

D. Performance Analysis

A comprehensive analysis of the trained models is conducted using standard evaluation metrics. Classification reports, including precision, recall, F1-score, and support, are generated to assess the performance of each model. Confusion matrices are also employed to visualize the classification results and gain deeper insights into the models' behavior, particularly in terms of correctly and incorrectly classified instances.

E. Hyperparameter Optimization

To further enhance the performance and robustness of the selected models, we perform hyperparameter optimization. This involves tuning the parameters of each model to find the optimal configuration that maximizes performance on the validation set and improves generalization to unseen data.

VI. SYSTEM ARCHITECTURE

Our implementation follows the steps outlined below and illustrated in Figure 1.

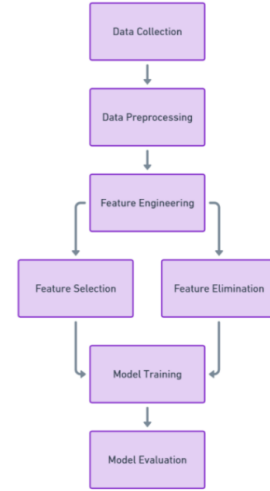


Fig. 1. System Architecture

The system architecture comprises the following key stages:

A. Data Collection

The ADS2018 dataset, containing real-world network traffic records, is used for training and evaluation.

B. Data Preprocessing

- Irrelevant attributes, such as IP addresses and timestamps, are removed. Missing and infinite values are handled to ensure data consistency.
- Feature engineering techniques, including standardization and categorical encoding, are applied to enhance model performance.

C. Data Analysis

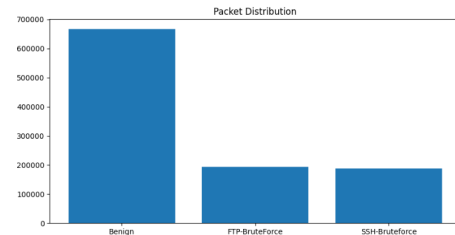


Fig. 2. Packet Distribution (Benign, FTP-Bruteforce, SSH-Bruteforce)

Figure 2 illustrates the class distribution for Benign, FTP-Bruteforce, and SSH-Bruteforce traffic. The Benign class significantly outnumbers the attack classes. While the FTP-Bruteforce and SSH-Bruteforce classes have similar packet counts, they represent much smaller proportions of the overall dataset.

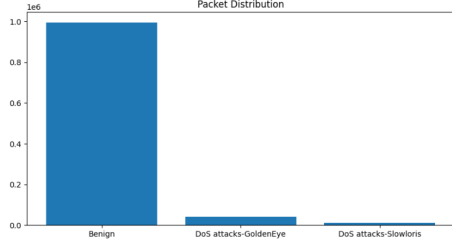


Fig. 3. Packet Distribution (Benign, DoS-GoldenEye, DoS-Slowloris)

Figure 3 shows the class distribution for Benign, DoS-GoldenEye, and DoS-Slowloris traffic. Again, the Benign class dominates, with over one million packets. The DoS attack categories have substantially fewer packets, with GoldenEye slightly more prevalent than Slowloris. Both DoS attack classes are significantly underrepresented compared to the Benign class.

D. Feature Selection

- Low-variance and highly correlated features are eliminated to reduce redundancy.
- Recursive Feature Elimination (RFE) is employed to identify and retain the most significant attributes.

E. Model Training

Various supervised learning models are implemented for classification. Performance is assessed using cross-validation to ensure robustness.

F. Model Evaluation

- Confusion matrices and precision-recall curves are generated for performance analysis.
- Feature importance graphs are used to interpret model decisions.

VII. RESULTS

The results in Table I compare the performance of five machine learning models: Random Forest, XGBoost, LightGBM, Extra Trees, and AdaBoost.

TABLE I
COMPARISON OF MACHINE LEARNING MODELS

Model	Train Acc.	Test Acc.	Prec.	Rec.	F1
Random Forest	94.41%	89.49%	0.90	0.89	0.89
XGBoost	91.60%	89.88%	0.90	0.89	0.89
LightGBM	92.38%	91.04%	0.91	0.91	0.91
Extra Trees	94.41%	89.42%	0.90	0.89	0.89
AdaBoost	31.53%	31.40%	0.18	0.31	0.19

Among these, LightGBM achieves the highest test accuracy (91.04%), along with the best F1-Score (0.91), indicating its effectiveness in balancing precision and recall. Random Forest and Extra Trees exhibit the same training accuracy (94.41%) but slightly lower test accuracy (89.49% and 89.42%, respectively), suggesting potential overfitting. XGBoost performs well with a test accuracy of 89.88% and an F1-Score of 0.89. In contrast, AdaBoost demonstrates poor performance across all metrics, with significantly lower training and test accuracies (31.53% and 31.40%) and an F1-Score of just 0.19, highlighting its unsuitability for this task. Overall, LightGBM emerges as the most effective model in this analysis.

VIII. CONCLUSION

Our study demonstrates that machine learning models can significantly enhance anomaly detection capabilities. LightGBM outperformed other models, achieving the best balance of accuracy and efficiency. Future work should explore deep learning techniques, ensemble methods, and real-time ADS deployment to further improve cybersecurity.

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